

Topology General Exam – January 11, 2024

Student ID: _____

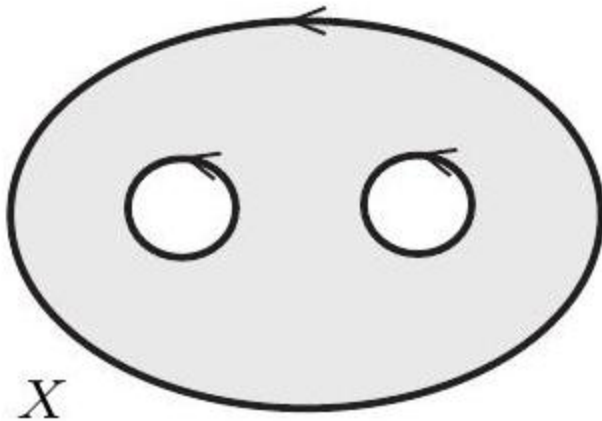
Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style. All work should be your own-no outside sources are permitted-using methods and results from the first year topology course topics. There are 8 problems; each problem is worth the same number of points.

Problem 1

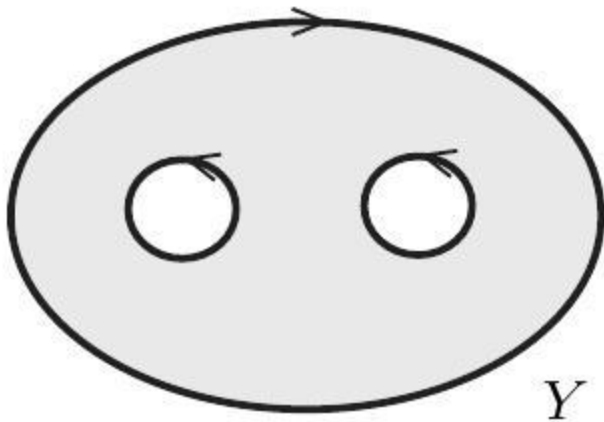
Let $f, g : \mathbb{R}^1 \rightarrow \mathbb{R}^2$ be two smooth maps. Prove that the subset of $\mathbb{R}^1 \times \mathbb{R}^1 \times \mathbb{R}^2$ consisting of all points $(x, y, z) \in \mathbb{R}^1 \times \mathbb{R}^1 \times \mathbb{R}^2$ satisfying the equation $f(x) - g(x) = z$ is a submanifold. What is its dimension?

Problem 2

Consider two topological spaces X, Y obtained from a disk by removing two smaller disks from its interior, and then identifying all three boundary curves to a single curve, respecting the orientations as shown in the figure. Compute the homology groups of both X and Y with $\mathbb{Z}$ coefficients.



For X , the outer boundary of the disk and both inner boundary circles are all oriented counterclockwise. All three boundary circles are identified to one curve respecting these arrows.



For Y , the outer boundary of the disk is oriented clockwise, while both inner boundary circles are oriented counterclockwise. All three boundary circles are identified to one curve respecting these arrows.

Problem 3

Show that any smooth map $f : \mathbb{R}P^n \rightarrow \mathbb{R}P^n$ has a fixed point, when n is even.

Problem 4

Prove that 1-forms $\theta_1, \dots, \theta_n$ on a smooth n -manifold M are linearly independent (thought of as sections of the cotangent bundle) if and only if $\theta_1 \wedge \dots \wedge \theta_n$ is non-vanishing.

Problem 5

Suppose the following diagram of abelian groups and homomorphisms commutes:

$$\begin{array}{ccccccc}
 & & & 0 & & 0 & \\
 & & & \downarrow & & \downarrow & \\
 & & & B_1 & \xrightarrow{\beta} & B_2 & \\
 & & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & A_1 & \longrightarrow & C & \longrightarrow & D \longrightarrow 0 \\
 & & \downarrow \alpha & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_2 & \longrightarrow & E & \longrightarrow & F \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

The rows are $0 \rightarrow A_1 \rightarrow C \rightarrow D \rightarrow 0$ and $0 \rightarrow A_2 \rightarrow E \rightarrow F \rightarrow 0$. The two columns are $0 \rightarrow B_1 \rightarrow C \rightarrow E \rightarrow 0$ and $0 \rightarrow B_2 \rightarrow D \rightarrow F \rightarrow 0$, directed downward. The other vertical map is α from A_1 to A_2 , and the top horizontal map is β from B_1 to B_2 . Every square commutes.

Assume that the rows and columns form exact sequences. Prove that there are isomorphisms $\ker \alpha \cong \ker \beta$ and $\operatorname{coker} \alpha \cong \operatorname{coker} \beta$.

Problem 6

Let $M = \{x^2 + y^2 + z^2 = 1\} \subset \mathbb{R}^3$, and for a real number $c \neq 0$ let $N = \{x^2 - y^2 + z^2 = c\}$. For which values of c is the intersection between M and N transverse? For which values of c is this intersection a submanifold of $\mathbb{R}^3$?

Problem 7

Let $\exp : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$ be the complex exponential map $\exp(z) = e^z$ (recall that $\exp$ is a universal covering map). If X is a connected, locally path-connected space and $f : X \rightarrow \mathbb{C} \setminus \{0\}$ a continuous map, a logarithm of f is a continuous map $g : X \rightarrow \mathbb{C}$ such that $\exp \circ g = f$. Prove that if such an X has finite fundamental group, then any continuous map $f : X \rightarrow \mathbb{C} \setminus \{0\}$ has a logarithm.

Problem 8

The connected sum of the 2-torus T and the projective plane $\mathbb{R}P^2$ is a smooth closed 2-manifold M whose fundamental group has a presentation $\pi_1(M) = \langle a, b, c \mid aba^{-1}b^{-1}c^2 = 1 \rangle$. Determine the number of connected, regular 3-fold covering spaces of M up to equivalence.

Hint: First show that this number is the same as the number of surjections $\pi_1(M) \rightarrow \mathbb{Z}/3\mathbb{Z}$ up to equivalence, where two surjections f, g are equivalent if and only if $g = \phi \circ f$ for some automorphism ϕ of $\mathbb{Z}/3\mathbb{Z}$.